Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 4 · SLO 2 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 4 : SLO â€“ 2 | Implementation of Structures Using Pointers â€” Fill in the Blanks

I. Fill in the Blanks

1. * (asterisk) â€” a structure pointer is declared using the * operator: struct S *ptr;
2. -> (arrow) â€” the arrow operator is used to access members via a pointer.
3. the address â€” a structure pointer stores the address of a structure variable.
4. &var â€” ptr = &var; assigns the address.
5. direct (efficient member) â€” pointers enable direct access to the structure's members without copying.
6. allocated (initialized) â€” memory for the pointed structure must be allocated before use.
7. in-place (permanent) â€” passing a pointer to a function lets it modify the original structure.
8. ptr->member â€” (*ptr).member is the same as ptr->member.
9. structure (data) â€” pointers reduce structure copying overhead.
10. malloc â€” dynamic allocation: ptr = malloc(sizeof(struct S));
11. size (amount of memory) â€” sizeof(struct) ensures the correct allocation size.
12. freed â€” unused structure pointers' memory should be freed.
13. dynamic â€” pointers enable dynamic structures like linked lists.
14. dangling pointers (undefined behavior) â€” always initialize to NULL or a valid address.
15. member access (the dot) â€” the arrow operator combines dereference and member access.